



2012 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 1

Monday 27th February 2012

General Instructions

- Writing time — 2 Hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Collection

Section I Questions 1–10

- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.

Section II Questions 11–14

- Start each of these questions in a new booklet.
- Write your name, class and master clearly on each booklet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Candidature — 128 boys

Examiner
MLS

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

Question One

If $f(x) = \frac{x-1}{x}$, which of the following is equal to $f\left(\frac{1}{a}\right)$?

1

(A) $1 - a$

(B) $\frac{a}{a-1}$

(C) $1 + a$

(D) $\frac{a-1}{a}$

Question Two

Which of the following is the solution to the inequation $\frac{x-3}{x} \leq 0$?

1

(A) $x \leq 3$

(B) $x < 0$ or $x \geq 3$

(C) $0 < x \leq 3$

(D) $0 \leq x \leq 3$

Question Three

Which of the following is the derivative of $2 \sin^{-1} 5x$?

1

(A) $\frac{10}{\sqrt{1-25x^2}}$

(B) $\frac{1}{\sqrt{1-25x^2}}$

(C) $\frac{5}{\sqrt{1-25x^2}}$

(D) $\frac{10}{\sqrt{25-x^2}}$

Exam continues next page ...

Question Four

The area under the curve $y = \frac{1}{x}$ between $x = 1$ and $x = a$ is 1 square unit.

1

What is the value of a ?

- (A) e
- (B) 0
- (C) $\ln 2$
- (D) 1

Question Five

The acute angle between the lines $y = 2x - 5$ and $y = 5x + 3$ is α .

1

What is the value of $\tan \alpha$?

- (A) $\frac{3}{11}$
- (B) $-\frac{3}{11}$
- (C) $\frac{7}{9}$
- (D) $-\frac{7}{9}$

Question Six

Suppose A is the point $(1, -2)$ and B is the point $(5, 6)$. The point $P(9, 14)$ divides the interval AB externally in what ratio?

1

- (A) $1:2$
- (B) $1:1$
- (C) $3:1$
- (D) $2:1$

Question Seven

What is the domain of $y = \sin^{-1} 2x$?

1

(A) $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

(B) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

(C) $-2 \leq x \leq 2$

(D) $-\frac{1}{2} \leq x \leq \frac{1}{2}$

Question Eight

What is an expression for $\int \frac{dx}{16 + x^2}$?

1

(A) $\frac{1}{4} \tan^{-1} 4x + c$

(B) $4 \tan^{-1} \frac{x}{4} + c$

(C) $4 \tan^{-1} 4x + c$

(D) $\frac{1}{4} \tan^{-1} \frac{x}{4} + c$

Question Nine

What is the Cartesian equation of the curve $x = 2 \sin \theta, y = 2 \cos \theta$?

1

(A) $x^2 + y^2 = \sqrt{2}$

(B) $x^2 + y^2 = 4$

(C) $x^2 = 4y$

(D) $y^2 = 4x$

Question Ten

Which of the following functions is a primitive of $\sin^2 x$?

1

(A) $\frac{1}{2}x - \frac{1}{4}\sin x$

(B) $\frac{1}{2}x - \frac{1}{4}\sin 2x$

(C) $\frac{1}{2}x - \frac{1}{4}\cos x$

(D) $\frac{1}{2}x - \frac{1}{4}\cos 2x$

————— End of Section I —————

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

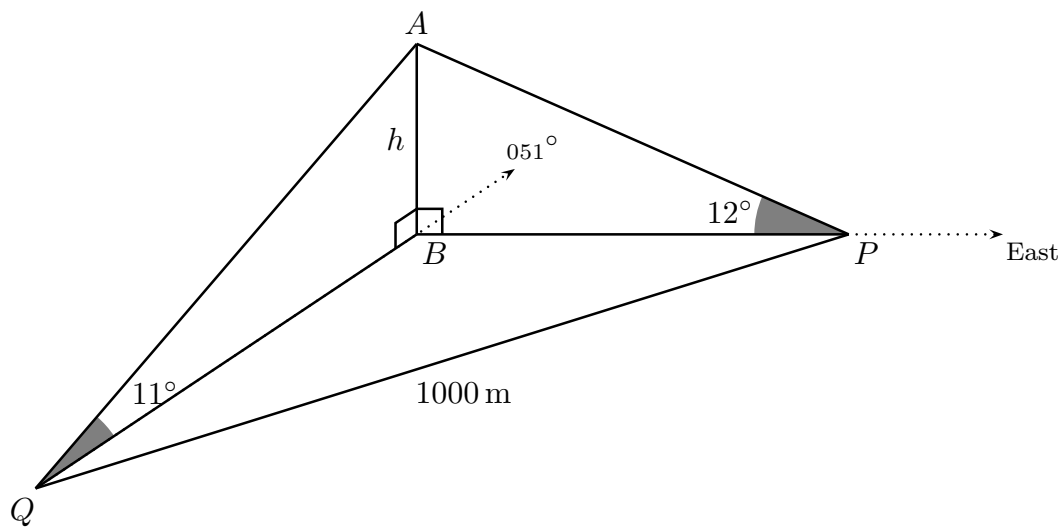
Start a new booklet for each question.

Question Eleven (15 marks) Use a separate writing booklet.	Marks
(a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.	1
(b) Find the exact value of $\sin(\cos^{-1}(-\frac{3}{5}))$.	1
(c) Evaluate $\int_0^1 \frac{-1}{\sqrt{2-x^2}} dx$.	2
(d) Solve the equation $2\sin^2 \theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$.	2
(e) (i) Expand $\sin(A - B)$.	1
(ii) Prove that $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$.	3
(f) Find the volume of the solid formed when the region bounded by the parabola $y = 4 - x^2$ and the x -axis is rotated about the y -axis.	2
(g) The volume of a sphere is increasing at a constant rate of $200 \text{ cm}^3/\text{s}$. You are given that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$. Find the rate of change of the radius, $\frac{dr}{dt}$, when $r = 10 \text{ cm}$. Leave your answer in exact form.	3

Question Twelve (15 marks) Use a separate writing booklet.

Marks

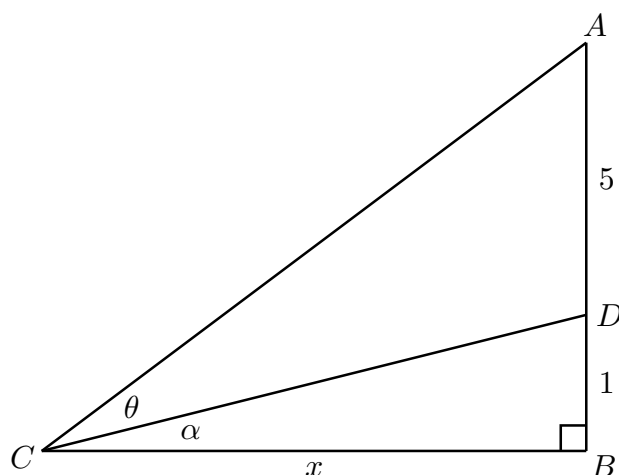
- (a) (i) Express $\sqrt{3}\cos x - \sin x$ in the form $A\cos(x + \alpha)$, where $A > 0$ and $0 \leq \alpha < 2\pi$. **2**
- (ii) Write down the maximum value of $\sqrt{3}\cos x - \sin x$. **1**
- (iii) Solve the equation $\sqrt{3}\cos x - \sin x = 1$, for $0 \leq x \leq 2\pi$. **2**
- (b)



The angle of elevation of a mobile phone tower AB of height h metres from a point P due east of the tower is 12° . From another point Q , the bearing of the mobile phone tower is 051° and the angle of elevation is 11° . The points P and Q are 1000 metres apart and on the same level as the base B of the tower.

- (i) Show that $\angle PBQ = 141^\circ$. **1**
- (ii) Show that $PB = h \tan 78^\circ$, and write a similar expression for QB . **1**
- (iii) Use the cosine rule in $\triangle PBQ$ to calculate h correct to the nearest metre. **2**

(c)



In the diagram above ABC is a triangle with a right angle at B . The point D lies on AB so that AD is 5 units and DB is 1 unit. Let CB be x units. The angle at C is divided into two angles marked θ and α as shown in the diagram.

- (i) Show that $\theta = \tan^{-1} \frac{6}{x} - \tan^{-1} \frac{1}{x}$. 1
- (ii) Show that θ is a maximum when $x = \sqrt{6}$. 3
- (iii) Deduce that the maximum size of $\angle ACD$ is $\theta = \tan^{-1} \frac{5\sqrt{6}}{12}$. 2

Question Thirteen (15 marks) Use a separate writing booklet.

Marks

(a) Consider the function $f(x) = 2 \tan^{-1} x$.

(i) Evaluate $f(\sqrt{3})$.

1

(ii) Draw the graph of $y = f(x)$, labelling any key features.

2

(b) Consider the function $f(x) = \frac{e^x}{5 + e^x}$.

(i) Show that $f(x)$ has no stationary points.

2

(ii) Show that $(\ln 5, \frac{1}{2})$ is a point of inflexion.

3

(iii) Find the domain and range of $f(x)$.

1

(iv) Sketch the curve $f(x) = \frac{e^x}{5 + e^x}$, showing any intercepts, asymptotes and points of inflexion.

3

(v) Explain why $f(x)$ has an inverse function.

1

(vi) Find the equation of the inverse function $y = f^{-1}(x)$.

1

(vii) State the domain and range of $y = f^{-1}(x)$.

1

Question Fourteen (15 marks) Use a separate writing booklet.

Marks

(a) Find the general solution of $\cos 2x + 3 \sin x = 2$.

4

(b) (i) By considering the sum of an arithmetic series, show that

1

$$(1 + 2 + 3 + \dots + n)^2 = \frac{1}{4}n^2(n + 1)^2.$$

(ii) By using the Principle of Mathematical Induction prove that

4

$$1^3 + 2^3 + 3^3 + \dots + n^3 = (1 + 2 + 3 + \dots + n)^2$$

for all integers $n \geq 1$.

(c) Two distinct points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. You are given $p > q > 0$.

(i) Show that the equation of the tangent to the parabola at P is $y = px - ap^2$.

1

(ii) The tangents to the parabola at P and Q meet at T . Find the co-ordinates of T .

1

(iii) The tangents at P and Q intersect at an angle of 45° . Show that $p - q = 1 + pq$.

1

(iv) Find the equation of the locus of T given that the tangents at P and Q intersect at an angle of 45° .

3

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SYDNEY GRAMMAR SCHOOL



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

NAME:

CLASS: MASTER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

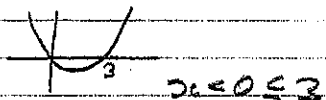
A ☐ B ☐ C ☐ D ☐

Solutions
Form VI Ext 1

Q1. $\frac{1}{a} = 1 \Rightarrow \frac{1-a}{1}$

A

Q2. $x(x-3) \leq 0$



C

Q3. $y = 2 \sin^{-1} 5x$
 $y' = 2 \times \frac{5}{\sqrt{1-25x^2}}$

A

Q4.



$$\int_1^a \frac{1}{x} dx = \ln x \Big|_1^a = 1$$

$$= \ln a - \ln 1 = 1$$

$$= \ln a = 1$$

$$a = e$$

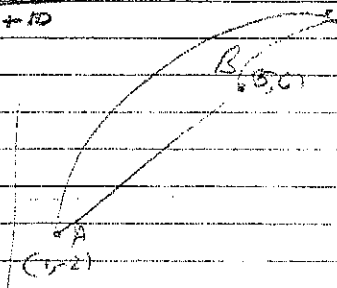
A

Q5.

$\tan \alpha = \frac{2-5}{1+10}$

A

Q6.



D

Q7.

$-1.522 \leq 1$
 $-3.522 \leq \frac{1}{2}$

D

Q8.

$a = 4$

D

Q9.

$2^2 + y^2 = 4 \sin^2 \theta + 4 \cos^2 \theta$
 $= 4$

B

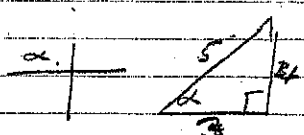
Q10

B

Q11.

(a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$
 $= 3$ ✓ (just need 3)

(b) let $\cos^{-1}(-\frac{3}{5}) = \alpha$, $0 \leq \alpha \leq \pi$
 $\sin \alpha = \frac{4}{5}$ ✓



(c) $\int_0^1 \frac{-1}{\sqrt{2-2x}} dx = \left[\cos^{-1} \frac{x}{\sqrt{2}} \right]_0^1$ ✓
 $= \cos^{-1} \frac{1}{\sqrt{2}} - \cos^{-1} 0$
 $= \frac{\pi}{4} - \frac{\pi}{2} = -\frac{\pi}{4}$ ✓
 must be in radians

(d) $2 \sin^2 \theta = \sin \theta$
 $2 \sin^2 \theta - \sin \theta = 0$
 $\sin \theta (2 \sin \theta - 1) = 0$ ✓
 $\sin \theta = 0$ or $\sin \theta = \frac{1}{2}$
 $\theta = 0, \pi, 2\pi$ or $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$
 $\theta = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}$ ✓ (need all for 2nd mk)

(e) (i) $\sin(A-B) = \sin A \cos B - \cos A \sin B$ ✓
 (ii) LHS = $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta}$
 $= \frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\sin \theta \cos \theta}$ ✓

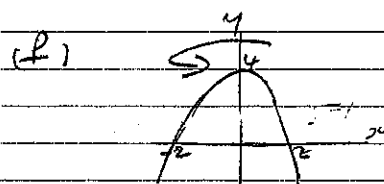
$$= \frac{\sin(3\theta - \theta)}{\sin \theta \cos \theta}$$

$$= \frac{\sin 2\theta}{\sin \theta \cos \theta}$$

$$= \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta}$$

$$= 2$$

$$= \text{RHS.}$$



$$V = \pi \int_{-2}^2 x^2 dy$$

 $= \pi \int_0^4 (4-y) dy$ ✓
 $= \pi \left[4y - \frac{1}{2}y^2 \right]_0^4$
 $= \pi (16 - 8) = 8\pi$ ✓

(g) Now $\frac{dr}{dt} = \frac{dr}{dV} \frac{dV}{dt}$ ✓

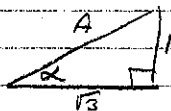
$V = \frac{4}{3}\pi r^3$	$\frac{dV}{dr} = 4\pi r^2$
$\frac{dr}{dt} = \frac{1}{4\pi r^2} \times 200$	$\frac{dr}{dV} = \frac{1}{4\pi r^2}$ ✓
$= \frac{1}{4\pi \times 100} \times 200$	
$= \frac{1}{2\pi} \text{ cm s}^{-1}$ ✓	

(There are other ways to do this)

Q12.

Q (i) $\sqrt{3} \cos x - \sin x = A \cos(x + \alpha)$
 $= A \cos x \cos \alpha - A \sin x \sin \alpha$

So $\sqrt{3} = A \cos \alpha$ and $1 = A \sin \alpha$



$\Rightarrow A^2 = 1 + 3 = 4$

$A = 2$

$\Rightarrow \tan \alpha = \frac{1}{\sqrt{3}}$
 $\alpha = \frac{\pi}{6}$

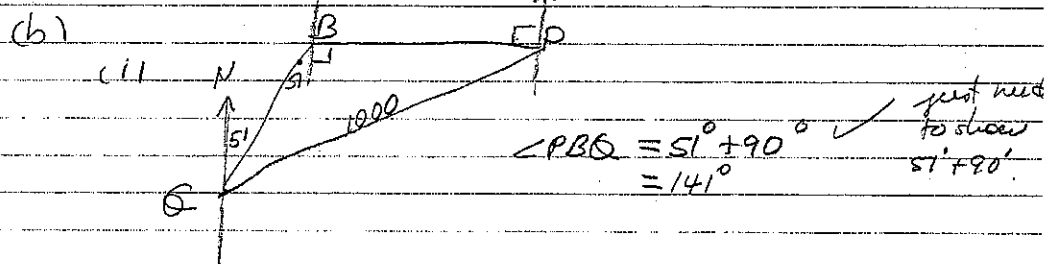
So $\sqrt{3} \cos x - \sin x = 2 \cos(x + \frac{\pi}{6})$

(i) 2 ✓

(iii) $\sqrt{3} \cos x - \sin x = 2 \cos(x + \frac{\pi}{6}) = 1$
 $\cos(x + \frac{\pi}{6}) = \frac{1}{2}$ ✓
 related angle is $\frac{\pi}{3}$

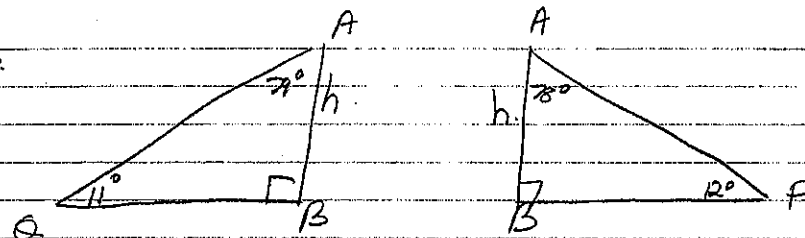
so $x + \frac{\pi}{6} = \frac{\pi}{3}$ or $\frac{5\pi}{3}$

$x = \frac{\pi}{6}$ or $\frac{3\pi}{2}$ ✓



(ii) over

ii



Using $\triangle ABP$, $\tan 78^\circ = \frac{BP}{h}$ ✓

so $BP = h \tan 78^\circ$

Similarly $QB = h \tan 79^\circ$ — if they don't write this have no matter they will have to use it in (iv)

iii

$QP^2 = QB^2 + BP^2 - 2 \times QB \times BP \times \cos 141^\circ$
 $1000^2 = h^2 \tan^2 79^\circ + h^2 \tan^2 78^\circ - 2 \times h^2 \tan 79^\circ \tan 78^\circ \times \cos 141^\circ$ ✓

$h^2 = \frac{1000^2}{\tan^2 79^\circ + \tan^2 78^\circ - 2 \tan 79^\circ \tan 78^\circ \cos 141^\circ}$

$= \frac{1000^2}{86.21855}$

$= 11598$

$h = 107.69$
 $\approx 108 \text{ m}$ ✓

(e)

$$(i) \angle ACD = \angle ACB - \angle DCB$$

$$\theta = \tan^{-1} \frac{6}{x} - \tan^{-1} \frac{1}{x}$$

we need them to identify $\angle ACB$ as $\tan^{-1} \frac{6}{x}$
and $\angle DCB$ as $\tan^{-1} \frac{1}{x}$ in some way.

(ii)

$$\theta = \tan^{-1} \frac{6}{x} - \tan^{-1} \frac{1}{x}$$

$$\frac{d\theta}{dx} = \frac{1}{1+\frac{36}{x^2}} \times \left(\frac{-6}{x^2} \right) - \frac{1}{1+\frac{1}{x^2}} \times \left(\frac{-1}{x^2} \right) = 0 \text{ at stat pt.}$$

$$\frac{-6}{x^2+36} + \frac{1}{1+x^2} = 0$$

$$\frac{6}{x^2+36} = \frac{1}{1+x^2}$$

$$6+6x^2 = x^2+36$$

$$5x^2 = 30$$

$$x^2 = 6$$

$$x = \pm \sqrt{6}$$

$$x = \sqrt{6}, \quad x > 0 \text{ since it is a length.}$$

check for maximum

x	$\sqrt{5}$	$\sqrt{6}$	$\sqrt{7}$
$\frac{d\theta}{dx}$	$\frac{-6}{41} + \frac{1}{6}$	0	$\frac{-6}{43} + \frac{1}{8}$
$\frac{d^2\theta}{dx^2}$	0.02 +ve		-0.01 -ve

So we have max θ for $x = \sqrt{6}$

$$(iii) \quad x = \sqrt{6}$$

$$\theta = \tan^{-1} \frac{6}{\sqrt{6}} - \tan^{-1} \frac{1}{\sqrt{6}}$$

$$\tan \theta = \frac{\frac{6}{\sqrt{6}} - \frac{1}{\sqrt{6}}}{1 + \frac{6}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}}}$$

$$= \frac{\frac{5}{\sqrt{6}}}{1+1}$$

$$= \frac{5}{2\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}}$$

$$= \frac{5\sqrt{6}}{12}$$

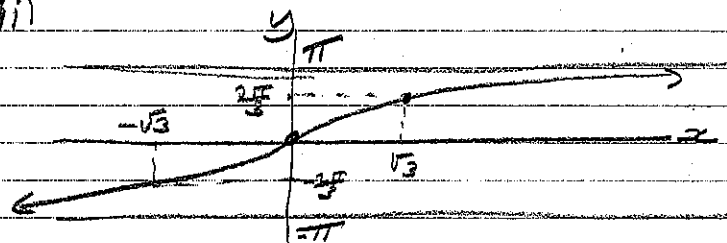
$$\text{so } \theta = \tan^{-1} \frac{5\sqrt{6}}{12}$$

Q13.

a)

(i) $f(\sqrt{3}) = 2 + \tan^{-1} \sqrt{3}$
 $= \frac{2\pi}{3}$ ✓

ii)



✓ asymptotes
 ✓ shape.

b) (i)

$$f(x) = \frac{e^x}{5+e^x}$$

$$f'(x) = \frac{(5+e^x)e^x - e^x e^x}{(5+e^x)^2}$$
 ✓

$$= \frac{se^x}{(5+e^x)^2} \neq 0 \text{ since } se^x \neq 0.$$

So no stationary points

(ii) $f''(x) = \frac{(5+e^x)^2 se^x - se^x \cdot 2(5+e^x)e^x}{(5+e^x)^4}$
 $= \frac{se^x(5+e^x)[5+e^x - 2e^x]}{(5+e^x)^4}$
 $= \frac{se^x(5+e^x)(5-e^x)}{(5+e^x)^4}$ ✓
 $= \frac{se^x(5-e^x)}{(5+e^x)^3}$ ✓

$f'(x) = 0$ at a possible pt of inflection

$$se^x(5-e^x) = 0$$

$$e^x = 5$$
 ✓

$$x = \ln 5$$

$$y = \frac{e}{5+e^{\ln 5}}$$

$$= \frac{5}{10}$$

$$= \frac{1}{2}$$

Sub for concavity change

x	$\ln 4$	$\ln 5$	$\ln 6$
$f''(x)$	20.9.1 9° +ve	0	30.11.(-1) 11° -ve

We have concavity change

So, $(\ln 5, \frac{1}{2})$ is a point of inflection

(iii)

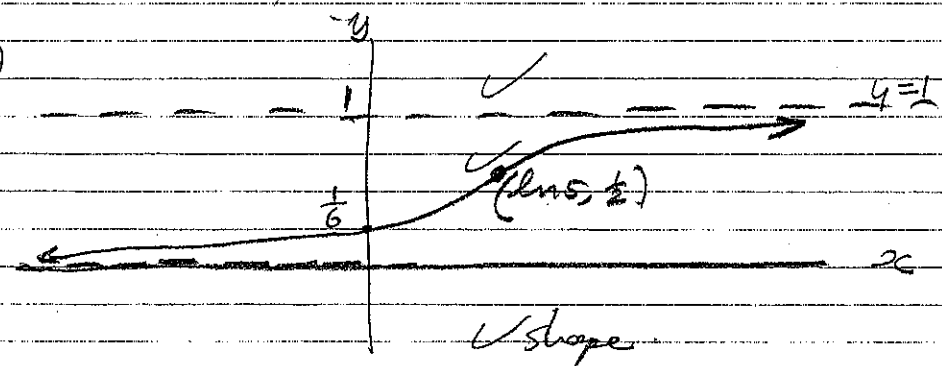
D: all x

R: $0 \leq y \leq 1$

✓

need both for marks

(iv)



(v) $f(x)$ has an inverse because a horizontal line cuts it once only. ✓
 (or any good reason) e.g. the function is increasing for all x .

(vi) $y = \frac{e^x}{5+e^x}$

$$x = \frac{e^y}{5+e^y}$$

$$5x + xe^y = e^y$$

$$e^y(1-x) = 5x$$

$$e^y = \frac{5x}{1-x}$$

$$y = \ln\left(\frac{5x}{1-x}\right) \quad \checkmark$$

(vii) D: $0 < x < 1$

R: all y . ✓

✓ If (vii) corresponds with (iii)

Q14.

(a) $\cos 2x + 3\sin x = 2$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x$$

$$1 - 2\sin^2 x + 3\sin x = 2 \quad \checkmark$$

$$2\sin^2 x - 3\sin x + 1 = 0$$

$$(2\sin x - 1)(\sin x - 1) = 0 \quad \checkmark$$

$$\sin x = \frac{1}{2}$$

$$\text{or } \sin x = 1$$

related angle is $\frac{\pi}{6}$ ✓

$$x = 2\pi + \frac{\pi}{2} \quad \checkmark$$

$$x = \pi + (-1)^n \frac{\pi}{6}, \text{ non integer}$$

✓

Note there are many other correct ways to express these answers.
 Accept answers in degrees.

(b) (i) $1+2+3+\dots+n = \frac{n}{2}(1+n) \quad \checkmark$

$$\text{so } (1+2+3+\dots+n)^2 = \frac{n^2}{4}(1+n)^2$$

(ii)

A: Consider $n=1$.

$$\text{LHS} = 1^3 = 1$$

$$\text{RHS} = 1^2 = 1 \quad \checkmark$$

So the statement is true for $n=1$.

B: Suppose the statement is true for some integer k , $k > 1$.

$$\text{i.e. suppose } 1^3 + 2^3 + \dots + k^3 = (1+2+\dots+k)^2$$

and show that $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = (1+2+\dots+k+(k+1))^2$

Now $1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3$

$$= (1+2+3+\dots+k)^2 + (k+1)^2 \quad \text{using the induction hypothesis}$$

$$= \frac{1}{4}k^2(k+1)^2 + (k+1)^2 \quad \text{using (i) } \checkmark$$

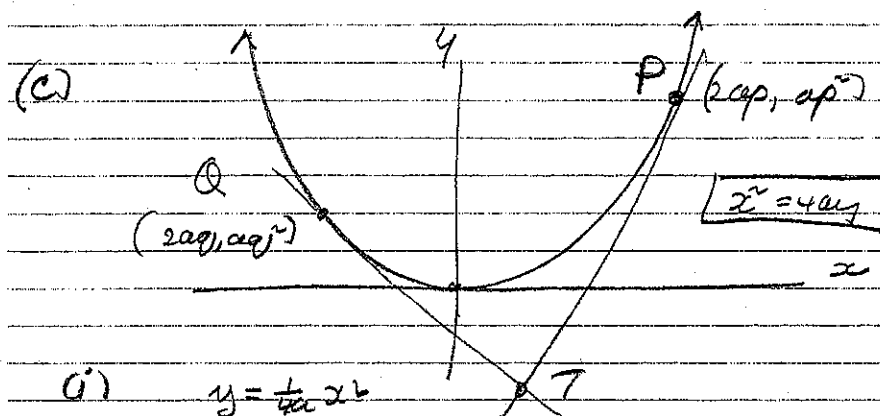
$$= \frac{(k+1)^2}{4} (k^2 + 4k + 4)$$

$$= \frac{1}{4} (k+1)^2 (k+2)^2 \quad \checkmark$$

$$= \frac{1}{4} (k+1)^2 ((k+1)+1)$$

$$= (1+2+3+\dots+(k+1))^2 \quad \text{using (i)}$$

C: So, by step A+B and Mathematical induction the given statement is true
(need the last statement for full marks).



(i) $y = \frac{1}{4a}x^2$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$x = 2ap, \quad m = \frac{2ap}{2a} = p$$

So tangent is $y - ap^2 = p(x - 2ap)$
 $y = px - ap^2$

(ii) $y = px - ap^2$
 $y = qx - aq^2$

$$qx - aq^2 = px - ap^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p-q) = a(p+q)(p-q) \quad p \neq q$$

$$x = a(p+q)$$

$$y = ap(p+q) - ap^2 \quad \checkmark$$

$$y = aq^2$$

T is $(a(p+q), aq^2)$

$$(iii) \tan 45^\circ = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$1 = \frac{p-q}{1+pq}, \quad \checkmark \quad p > q > 0.$$

$$1 + pq = p - q.$$

(iv)

$$\text{At } T, \quad x = a(p+q), \quad y = apq$$

$$\frac{x}{a} = p+q, \quad \frac{y}{a} = pq$$

$$\text{now } (p-q)^2 = (p+q)^2 - 4pq. \quad \checkmark$$

$$\text{so } (1+pq)^2 = \frac{x^2}{a^2} - \frac{4y}{a} \quad (\text{using iii}) \quad \checkmark$$

$$\left(1 + \frac{y}{a}\right)^2 = \frac{x^2}{a^2} - \frac{4y}{a}$$

$$(a+y)^2 = x^2 - 4ay \quad \checkmark$$

$$a^2 + 2ay + y^2 = x^2 - 4ay$$

$$\text{locus is } a^2 + 6ay + y^2 - x^2 = 0$$